Homework 8

EE 999 — Introduction to Control Systems

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Due Wednesday, November 11, 2026

Problem A — PI and PD Control

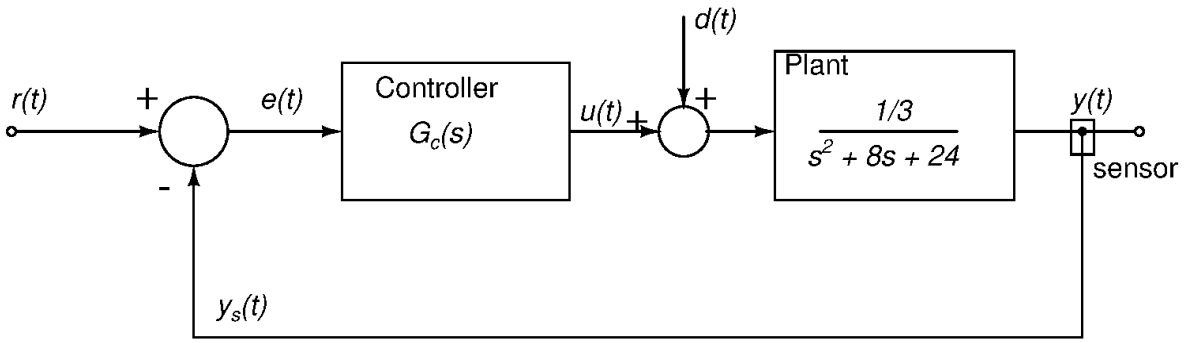


Figure 1: A system with PI-type control.

Problem A.1 — PI Control

Use $G_c(s) = \dfrac{k_p s + k_i}{s}$ *with* $k_p = 10$, $k_i = 200$.

Problem A.1.a

Determine the transfer functions:

i) $T_{ry}(s)$**, the input-output transfer function**

$$G_p(s) = \frac{1/3}{s^2 + 8s + 24} \tag{1}$$

$$G_c(s) = \frac{k_p s + k_i}{s} \tag{2}$$

$$Y(s) = \frac{G_p G_c}{1 + G_p G_c} R(s) \tag{3}$$

$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} R(s) \tag{4}$$

$$Y(s) = \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} R(s) \tag{5}$$

$$Y(s) = \frac{10s + 200}{3s(s^2 + 8s + 24) + 10s + 200} R(s) \tag{6}$$

$$Y(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{7}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

ii) $T_{re}(s)$**, the error transfer function**

$$E(s) = R(s) - Y(s) \tag{9}$$
$$E(s) = R(s) - T_{ry}(s)R(s) \tag{10}$$
$$E(s) = \left(1 - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{11}$$
$$E(s) = \left(\frac{3s^3 + 24s^2 + 82s + 200}{3s^3 + 24s^2 + 82s + 200} - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{12}$$
$$E(s) = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{13}$$
$$T_{re}(s) = \frac{E(s)}{R(s)} = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} \tag{14}$$

iii) $T_{dy}(s)$, the disturbance transfer function

$$Y(s) = G_p \left(D(s) + U(s)\right) \tag{15}$$
$$Y(s) = G_p \left(D(s) + G_c(s)E(s)\right) \tag{16}$$
$$Y(s) = G_p \left(D(s) - G_c(s)Y(s)\right) \tag{17}$$
$$Y(s) = G_p D(s) - G_p G_c(s) Y(s) \tag{18}$$
$$Y(s) + G_p G_c(s) Y(s) = G_p D(s) \tag{19}$$
$$Y(s) \left(1 + G_p G_c(s)\right) = G_p D(s) \tag{20}$$
$$Y(s) = \frac{G_p}{1 + G_p G_c(s)} D(s) \tag{21}$$
$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} D(s) \tag{22}$$
$$Y(s) = \frac{\frac{1}{3}s}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} D(s) \tag{23}$$
$$Y(s) = \frac{s}{3s(s^2 + 8s + 24) + 10s + 200} D(s) \tag{24}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 72s + 10s + 200} D(s) \tag{25}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 82s + 200} D(s) \tag{26}$$
$$T_{dy}(s) = \frac{Y(s)}{D(s)} = \frac{s}{3s^3 + 24s^2 + 82s + 200} \tag{27}$$

Problem A.1.b

Draw a pole-zero diagram of $T_{ry}(s)$, label the dominant mode.

Code: `hw_08_A1b.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

pzmap(sys);

sys_poles = pole(sys)
sys_zeros = zero(sys)
```

Output: `output/hw_08_A1b.txt`

```
sys_poles =

  -5.2097 +      0i
  -1.3952 + 3.2940i
  -1.3952 - 3.2940i

sys_zeros = -20
```

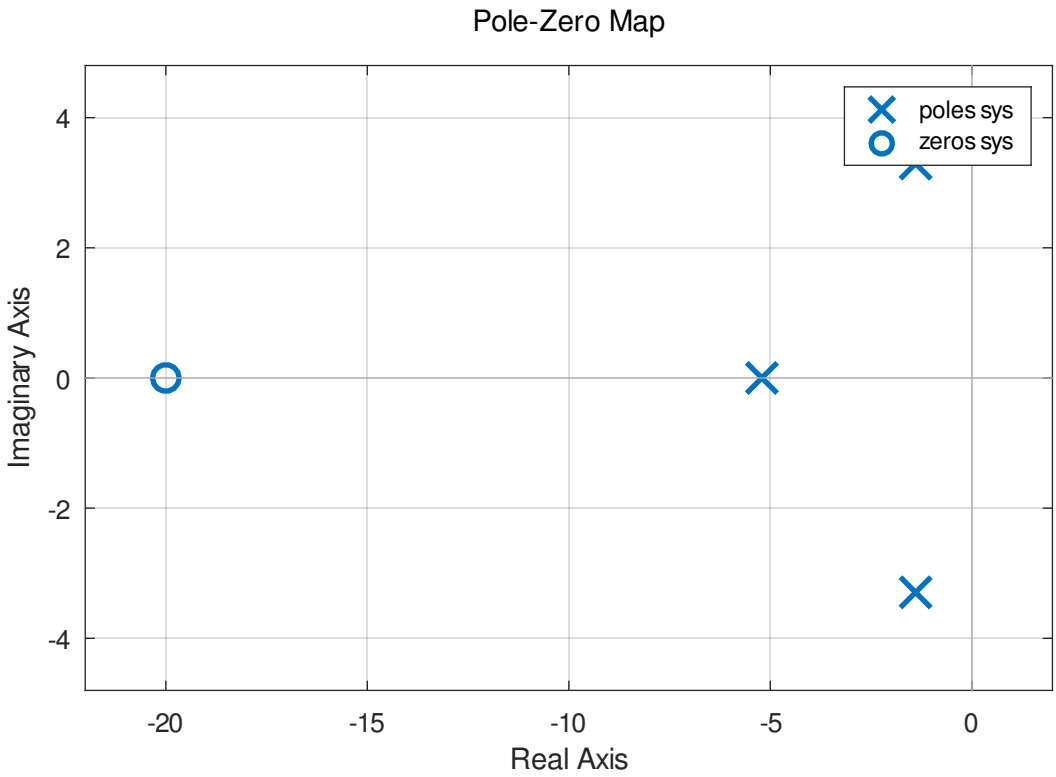


Figure 2: Pole-zero diagram of $T_{ry}(s)$.

The dominant poles are at $s = -1.3952 \pm 3.2940j$.

Problem A.2

Based on the dominant mode pole locations, determine

$$p_{1,2} = -\alpha \pm j\omega \tag{28}$$
$$= -1.3952 \pm 3.2940j \tag{29}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{30}$$
$$= \sqrt{1.3952^2 + 3.2940^2} \tag{31}$$
$$= 3.5773 \text{ rad/s} \tag{32}$$

i) t_p, the peak time

$$t_p = \frac{\pi}{\omega} \tag{33}$$
$$= \frac{\pi}{3.2940} \tag{34}$$
$$= 0.9537 \text{ s} \tag{35}$$

ii) ρ, the damping

$$\rho = \frac{\alpha}{\omega_n} \tag{36}$$
$$= \frac{1.3952}{3.5773} \tag{37}$$
$$= 0.3900 \tag{38}$$

iii) M_p, the percent overshoot

$$M_p = 100\% \times e^{-\pi\alpha/\omega} \tag{39}$$
$$= 100\% \times e^{-\pi(1.3952)/3.2940} \tag{40}$$
$$= 26.43\% \tag{41}$$

iv) t_r, the rise time

$$t_r \approx \frac{1.8}{\omega_n} \tag{42}$$
$$\approx \frac{1.8}{3.5773} \tag{43}$$
$$\approx 0.5032 \text{ s} \tag{44}$$

v) t_s, the settling time

$$t_s \approx \frac{4}{\alpha} \tag{45}$$
$$\approx \frac{4}{1.3952} \tag{46}$$
$$\approx 2.8670 \text{ s} \tag{47}$$

Problem A.3

Analytically determine e_{ss}, the steady state error, by determining the system type and K_{Bode}.

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \tag{48}$$
$$= \frac{10s + 200}{s} \cdot \frac{1/3}{s^2 + 8s + 24} \cdot 1 \tag{49}$$
$$= \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24)} \tag{50}$$

$G_L(s)$ has one pole at the origin, so the system is Type I. A Type I system has no steady-state error for a step input, $e_{ss} = 0$.

Problem A.4

Setup $T_{ry}(s)$ in Matlab, run and plot the (true) step response. Determine these values for the complete system

$$T_{ry}(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

Code: hw_08_A4.m

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

% T_ry(s) = (10s + 200) / (3s^3 + 24s^2 + 82s + 200)
num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

info = stepinfo(sys)

damp(sys)

step(sys);
grid on;
title('Step response of T_{ry}(s)');
```

Output: output/hw_08_A4.txt

```
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 0.4936
    TransientTime = 2.4754
    SettlingTime = 2.4754
    SettlingMin = 0.9000
    SettlingMax = 1.2038
    Overshoot = 20.376
    Undershoot = 0
    Peak = 1.2038
    PeakTime = 1.1429
    StaticGain = 1
    InitialJump = 0

   Pole                  Damping     Frequency       Time Constant
                                     (rad/seconds)   (seconds)

   -1.40e+00+3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -1.40e+00-3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -5.21e+00+0.00e+00i   1.00e+00    5.21e+00        1.92e-01
```

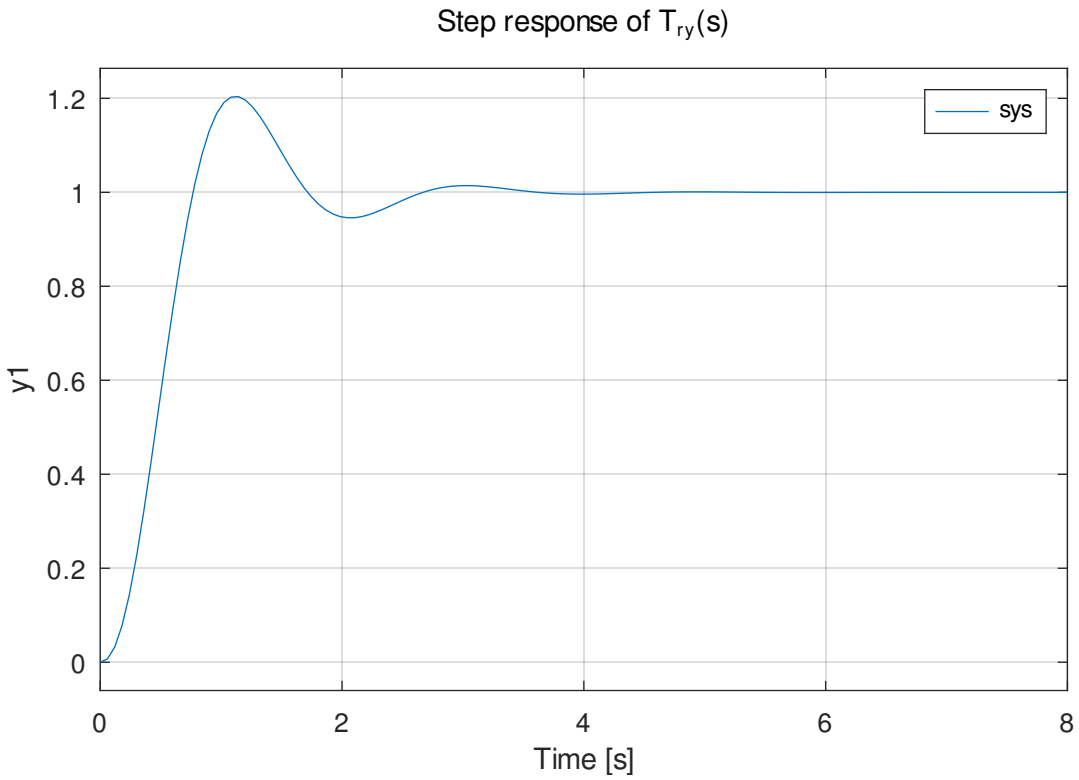


Figure 3: Step response of $T_{ry}(s)$.

i) t_p, the peak time

$$t_p = 1.1429 \text{ s} \tag{51}$$

ii) ρ, the damping

$$\rho = 0.3900 \tag{52}$$

iii) M_p, the percent overshoot

$$M_p = 20.38\% \tag{53}$$

iv) t_r, the rise time

$$t_r = 0.4936 \text{ s} \tag{54}$$

v) t_s, the settling time

$$t_s = 2.4754 \text{ s} \tag{55}$$

vi) e_{ss}, the steady state error

$$e_{ss} = 0 \tag{56}$$

Problem A.5

(Essay question) What is the value of the methods of steps A.1.b–A.2, relative to the method of step A.4?

The value of the methods in A.1.b–A.2 is that they provide reasonable estimates just by evaluating the dominant pole pair without the rigor of actually having to plot or otherwise analyze the entire system response. The tradeoff is that the numbers are not accurate.

Problem A.6

Based on parts A.1.b–A.4, which measures based on the dominant modes came out most accurate? Which measures based on the dominant modes (part 3) came out least accurate?

Measure	Estimate	Actual	Error
t_p, peak time	0.9537 s	1.1429 s	−16.6%
ρ, damping	0.3900	0.3900	0.0%
M_p, percent overshoot	26.43%	20.38%	+29.7%
t_r, rise time	0.5032 s	0.4936 s	+1.9%
t_s, settling time	2.8670 s	2.4754 s	+15.8%
e_{ss}, steady-state error	0	0	–

Table 1: Dominant-mode estimates (A.2, A.3) compared with the complete-system values (A.4). Error is (estimate − actual)/actual.

The damping and steady-state error were the most accurate, and the percent overshoot was the least accurate.

Problem B — PD Control

Use $G_c(s) = k_d s + k_p$ with $k_p = 300$, and k_d to be determined.

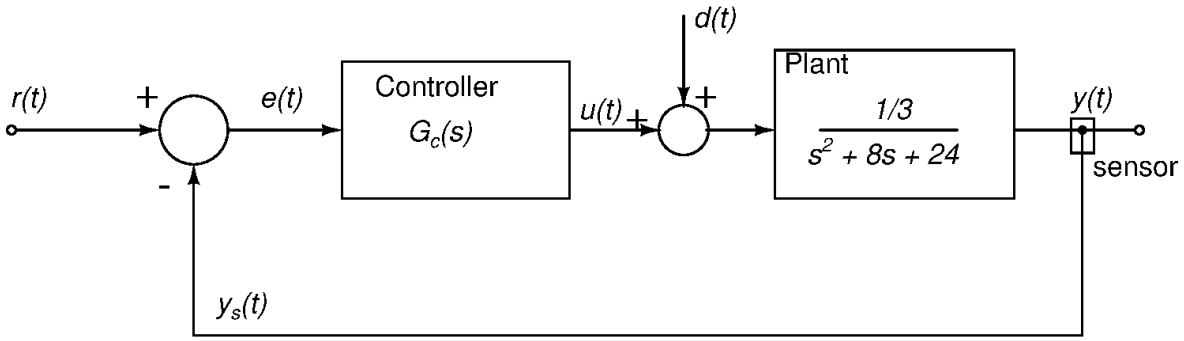


Figure 1: The system from Problem A, repeated, with PD control $G_c(s) = k_d s + k_p$ in place of PI.

Problem B.1 — Experiment with k_d

Problem B.1.a

Derive $T_{ry}(s)$, the closed loop transfer function with PD control, and with k_p and k_d as variables. Set $k_p = 300$ and determine the pole locations for several values of k_d on the range from 0.5 to 10.

$$G_c(s) = k_d s + k_p \tag{57}$$

$$G_p(s) = \frac{1/3}{s^2 + 8s + 24} \tag{58}$$

$$Y(s) = \frac{G_p G_c}{1 + G_p G_c} R(s) \tag{59}$$

$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24} \cdot (k_d s + k_p)}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot (k_d s + k_p)} R(s) \tag{60}$$

$$Y(s) = \frac{\frac{1}{3}(k_d s + k_p)}{s^2 + 8s + 24 + \frac{1}{3}(k_d s + k_p)} R(s) \tag{61}$$

$$Y(s) = \frac{k_d s + k_p}{3(s^2 + 8s + 24) + k_d s + k_p} R(s) \tag{62}$$

$$Y(s) = \frac{k_d s + k_p}{3s^2 + 24s + 72 + k_d s + k_p} R(s) \tag{63}$$

$$Y(s) = \frac{k_d s + k_p}{3s^2 + (24 + k_d)s + (72 + k_p)} R(s) \tag{64}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{k_d s + k_p}{3s^2 + (24 + k_d)s + (72 + k_p)} \tag{65}$$

Code: `hw_08_B1a.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

k_p = 300;

for k_d = [0.5 2 5 8 10]
    k_d

    num = [k_d k_p];
    den = [3 24+k_d 72+k_p];

    sys = tf(num, den);
    k_d_poles = pole(sys)
end
```

Output: `output/hw_08_B1a.txt`

```
k_d = 0.5000
k_d_poles =

   -4.083 + 10.360i
   -4.083 - 10.360i

k_d = 2
k_d_poles =

   -4.333 + 10.258i
   -4.333 - 10.258i

k_d = 5
k_d_poles =

   -4.833 + 10.032i
   -4.833 - 10.032i

k_d = 8
k_d_poles =

  -5.3333 + 9.7753i
  -5.3333 - 9.7753i

k_d = 10
k_d_poles =

  -5.6667 + 9.5859i
  -5.6667 - 9.5859i
```

Problem B.1.b

Hand sketch a plot of the closed-loop pole locations as a function of k_d.

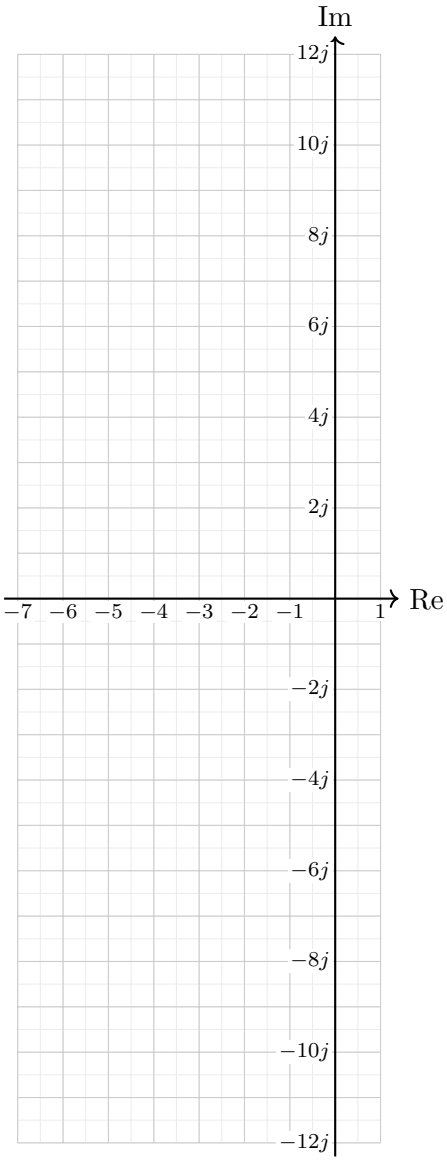


Figure 4: Closed-loop poles of $T_{ry}(s)$ as k_d varies from 0.5 to 10, with $k_p = 300$.

Problem B.1.c

Select a value of k_d to give $\rho = 0.4$, be rather accurate, e.g., $0.39 < \rho < 0.41$.

Code: `hw_08_B1c.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

k_p = 300;
rho_err = .1;

for k_d = 2:0.1:3
```

```
    num = [k_d k_p];
    den = [3 24+k_d 72+k_p];

    sys = tf(num, den);
    [w_n, rho, p] = damp(sys);

    rho_test = abs(0.4-rho(1));

    if rho_test < rho_err
        rho_err = rho_test;
        k_d_best = k_d;
        rho_best = rho(1);
        p_best = p;
    end
end

k_d_best
rho_best
p_best
```

Output: `output/hw_08_B1c.txt`

```
k_d_best = 2.7000
rho_best = 0.3996
p_best =

   -4.450 + 10.208i
   -4.450 - 10.208i
```

Using the value of k_d chosen to give $\rho = 0.4$:

Problem B.2

What is the dominant mode of this system? (Same as asking "where are the dominant pole(s)")

With $k_d = 2.7$:

$$p_{1,2} = -\alpha \pm j\omega \tag{66}$$
$$= -4.4500 \pm 10.2077j \tag{67}$$

Problem B.3

Analyze your system (analytically, based on the dominant mode) to determine:

$$p_{1,2} = -4.4500 \pm 10.2077j \tag{67}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{68}$$
$$= \sqrt{4.4500^2 + 10.2077^2} \tag{69}$$
$$= 11.1355 \text{ rad/s} \tag{70}$$

i) t_p, the peak time

$$t_p = \frac{\pi}{\omega} \tag{71}$$
$$= \frac{\pi}{10.2077} \tag{72}$$
$$= 0.3078 \text{ s} \tag{73}$$

ii) ρ, the damping

$$\rho = \frac{\alpha}{\omega_n} \tag{74}$$
$$= \frac{4.4500}{11.1355} \tag{75}$$
$$= 0.3996 \tag{76}$$

iii) M_p, the percent overshoot

$$M_p = 100\% \times e^{-\pi\alpha/\omega} \tag{77}$$
$$= 100\% \times e^{-\pi(4.4500)/10.2077} \tag{78}$$
$$= 25.42\% \tag{79}$$

iv) t_r, the rise time

$$t_r \approx \frac{1.8}{\omega_n} \tag{80}$$
$$\approx \frac{1.8}{11.1355} \tag{81}$$
$$\approx 0.1616 \text{ s} \tag{82}$$

v) t_s, the settling time

$$t_s \approx \frac{4}{\alpha} \tag{83}$$
$$\approx \frac{4}{4.4500} \tag{84}$$
$$\approx 0.8989 \text{ s} \tag{85}$$

Problem B.4

Analytically determine e_{ss}, the steady state error, by determining the system type and K_{Bode}.

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \tag{86}$$
$$= (k_d s + k_p) \cdot \frac{1/3}{s^2 + 8s + 24} \cdot 1 \tag{87}$$
$$= (2.7s + 300) \cdot \frac{1/3}{s^2 + 8s + 24} \cdot 1 \tag{88}$$
$$= \frac{0.9s + 100}{s^2 + 8s + 24} \tag{89}$$

$G_L(s)$ has no poles at the origin, so the system is Type 0. In Bode form,

$$G_L(s) = \frac{1}{s^{N_o}} K_{Bode} \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + 1}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + 1} \tag{90}$$
$$G_L(s) = \frac{100}{24} \cdot \frac{0.009s + 1}{\frac{1}{24}s^2 + \frac{1}{3}s + 1} \tag{91}$$

so $N_o = 0$ and $K_{Bode} = \frac{100}{24}$. For a Type 0 system with a step input,

$$e_{ss} = \frac{1}{1 + K_{Bode}} \tag{92}$$
$$= \frac{1}{1 + \frac{100}{24}} \tag{93}$$
$$= \frac{24}{124} \tag{94}$$
$$= 0.1935 \tag{95}$$

Problem B.5

Analyze the step response of your system (step response, full system, you can use the `step` *and* `stepinfo` *commands) to determine:*

From equation (65):

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{k_d s + k_p}{3s^2 + (24 + k_d)s + (72 + k_p)} \tag{65}$$

$$T_{ry}(s) = \frac{2.7s + 300}{3s^2 + 26.7s + 372} \tag{96}$$

Code: `hw_08_B5.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

% T_ry(s) = (2.7s + 300) / (3s^2 + 26.7s + 372)
num = [2.7 300];
den = [3 26.7 372];
sys = tf(num, den);

info = stepinfo(sys)

damp(sys)

step(sys);
grid on;
title('Step response of T_{ry}(s)');
```

Output: `output/hw_08_B5.txt`

```
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 0.1316
    TransientTime = 0.7463
    SettlingTime = 0.7463
    SettlingMin = 0.7258
    SettlingMax = 1.0113
    Overshoot = 25.403
    Undershoot = 0
    Peak = 1.0113
    PeakTime = 0.3084
    StaticGain = 0.8065
    InitialJump = 0

   Pole                  Damping     Frequency       Time Constant
                                     (rad/seconds)   (seconds)

   -4.45e+00+1.02e+01i   4.00e-01    1.11e+01        2.25e-01
   -4.45e+00-1.02e+01i   4.00e-01    1.11e+01        2.25e-01
```

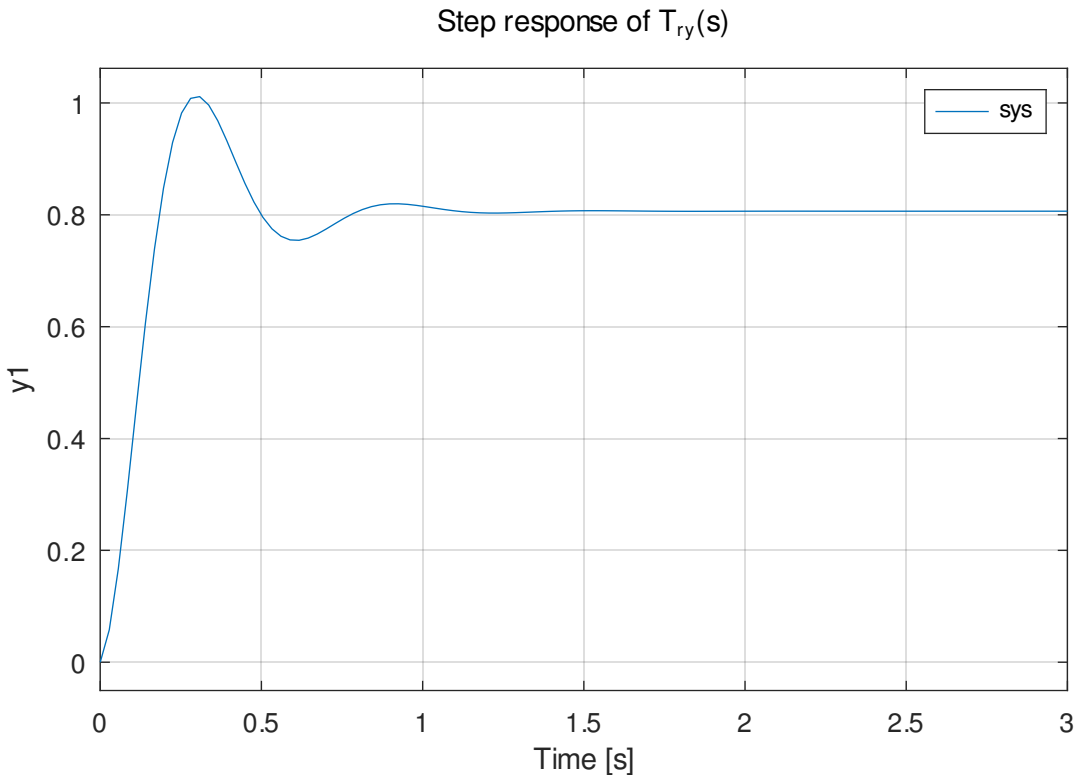


Figure 5: Step response of $T_{ry}(s)$ with PD control.

i) t_p, the peak time

$$t_p = 0.3084 \text{ s} \tag{97}$$

ii) ρ, the damping

$$\rho = 0.3996 \tag{98}$$

iii) M_p, the percent overshoot

$$M_p = 25.40\% \tag{99}$$

iv) t_r, the rise time

$$t_r = 0.1316 \text{ s} \tag{100}$$

v) t_s, the settling time

$$t_s = 0.7463 \text{ s} \tag{101}$$

vi) e_{ss}, the steady state error (not indicated by `stepinfo()`)

$$\begin{aligned} e_{ss} &= 1 - \text{StaticGain} && (102)\\ &= 1 - 0.8065 && (103)\\ &= 0.1935 && (104) \end{aligned}$$

Problem B.6

Which of the values based on dominant mode are closest to their analysis values? Which are farthest?

write the answer here

Problem C — Compare the PI and PD controllers

Consider these parameters; the desirable characteristic is indicated.

Controller performance measure	Desirable characteristic
i) t_p, the peak time	Fast
ii) ρ, the damping	Near one
iii) M_p, the percent overshoot	Near zero
iv) t_r, the rise time	Fast
v) t_s, the settling time	Fast
vi) e_{ss}, the steady-state error	Small

Problem C.1

What characteristic is most enhanced by the PI control system?

write the answer here

Problem C.2

What characteristic is most enhanced by the PD control system (tuned for $\rho = 0.4$)?

write the answer here